

Name _____

Period _____ Date _____

Practice Test – Advanced Kinematics (Version 6)

On any problems, you must show all work, include units, round properly, and box your final answer.

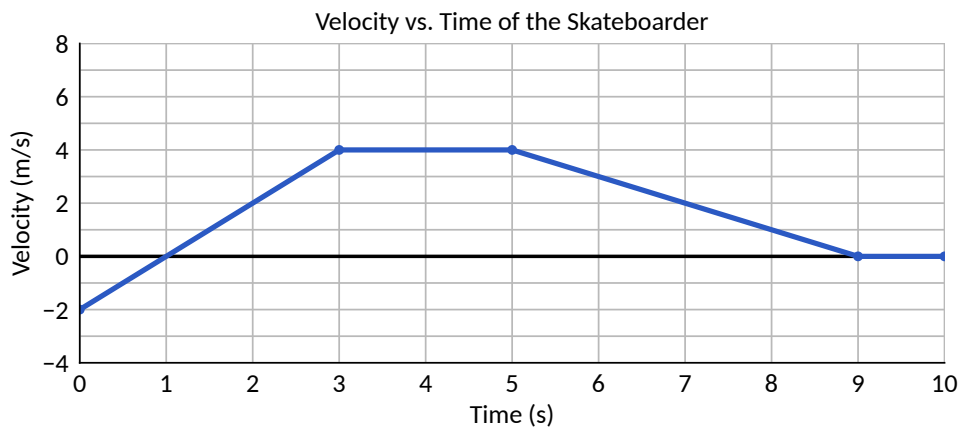
Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

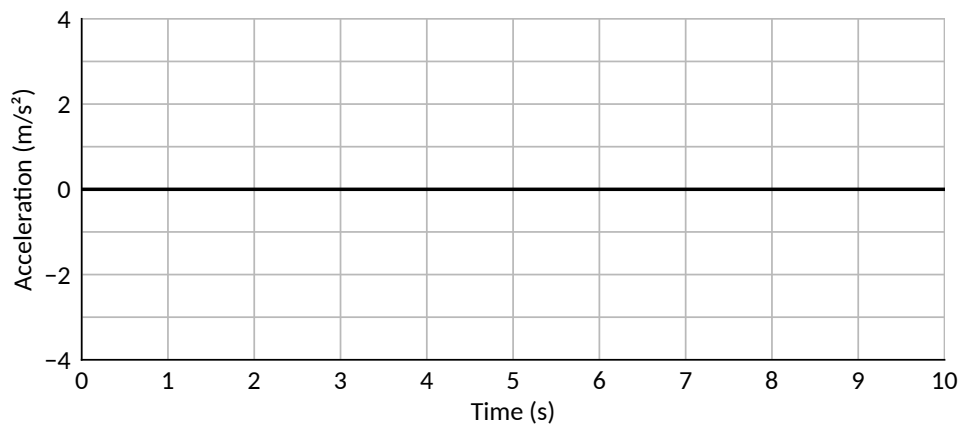
For all questions:

- Use $g = 9.80 \text{ m/s}^2$ and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).
- The last page is optional extra practice and is not part of the timed test.

1. (25 points) The velocity-time graph below shows the motion of a skateboarder rolling along a straight path from $t = 0$ s to $t = 10$ s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

2. (25 points) A train travels along a straight, level track at a constant 27.0 m/s. The engineer applies the brakes, and the train slows with a constant acceleration of magnitude 1.50 m/s^2 until it stops. How far does the train travel while braking?

3. (25 points) Water leaves a lawn sprinkler at ground level with a speed of 8.00 m/s at 60.0° above the horizontal and lands on the level lawn at the same height. Assume air resistance is negligible. How far from the sprinkler does the water land?

4. (25 points) A rock climber accidentally drops a carabiner from the top of a cliff that is a height L above the ground below. Assume air resistance is negligible. If down is the positive direction, what is the carabiner's velocity just before it hits the ground? Derive a mathematical expression in terms of L and g .

Optional extra practice (not part of the timed test, not scored)

E1. For the water in Question 3: (a) How long is it in the air? (b) What is its maximum height?

E2. For the train in Question 2: (a) How long does it take to stop? (b) If its initial speed were doubled and it slowed with the same acceleration, by what factor would its stopping distance change?

E3. For the carabiner in Question 4: write its velocity $v_y(t)$ as a function of the time t since it was released, and state the time interval over which this expression is valid.

Answer Key – Practice Test, Version 6 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units. $g = 9.80 \text{ m/s}^2$; answers to three significant figures.

AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):

#1 $v_x = v_{x0} + a_x t$ · #2 $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$ · #3 $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Same equations as $v_f = v_i + at$, $\Delta x = v_i t + \frac{1}{2} at^2$, $v_f^2 = v_i^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need. For vertical motion use y in place of x .

1. Acceleration-time graph of the skateboarder (25 points)

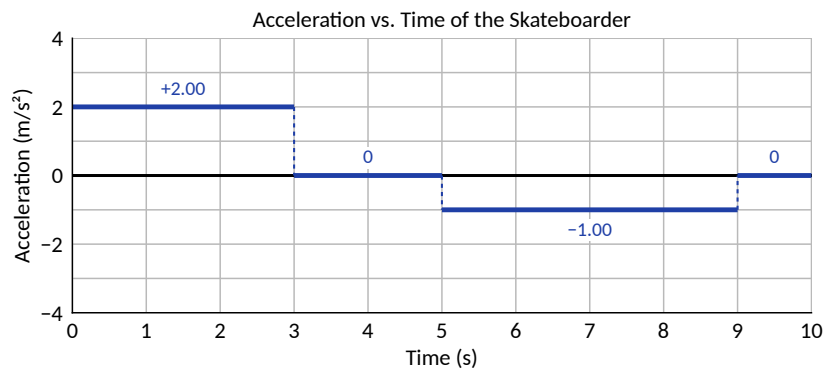
Step 1 Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece; values read off grid lines are treated as exact)}$$

Step 2 Find the slope of each segment.

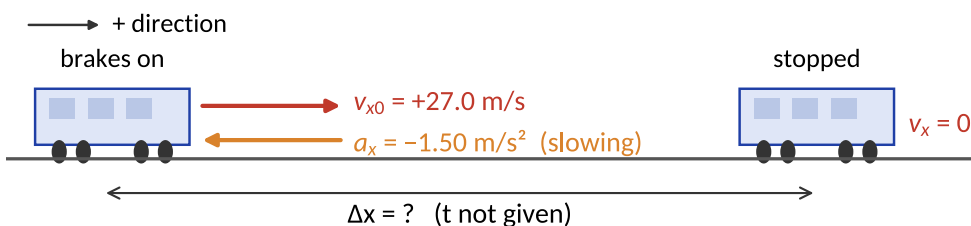
segment	$\Delta v = v_f - v_i$	$\Delta t = t_f - t_i$	$a = \Delta v / \Delta t$
0 s to 3 s	$4 \text{ m/s} - (-2 \text{ m/s}) = 6 \text{ m/s}$	$3 \text{ s} - 0 \text{ s} = 3 \text{ s}$	+2.00 m/s²
3 s to 5 s	$4 \text{ m/s} - 4 \text{ m/s} = 0 \text{ m/s}$	$5 \text{ s} - 3 \text{ s} = 2 \text{ s}$	0 m/s²
5 s to 9 s	$0 \text{ m/s} - 4 \text{ m/s} = -4 \text{ m/s}$	$9 \text{ s} - 5 \text{ s} = 4 \text{ s}$	-1.00 m/s²
9 s to 10 s	$0 \text{ m/s} - 0 \text{ m/s} = 0 \text{ m/s}$	$10 \text{ s} - 9 \text{ s} = 1 \text{ s}$	0 m/s²

Step 3 Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: from 0 s to 3 s the skateboarder starts out rolling backward ($v = -2 \text{ m/s}$), stops for an instant at 1 s, then rolls forward. The line is straight the whole time, so $a = +2.00 \text{ m/s}^2$ the whole time, even at the instant $v = 0$. Zero velocity does not mean zero acceleration.

2. Braking train: stopping distance (25 points)



Step 1 Pick a positive direction and get the sign of a.

Let the direction the train moves be +. The train is moving + but slowing down, so its velocity is decreasing: Δv is negative, which makes a negative. The slowing acceleration points opposite the motion, so $a = -1.50 \text{ m/s}^2$.

Step 2 List the five variables.

$v_i (= v_{x0})$	$v_f (= v_x)$	a	t	Δx
+27.0 m/s	0 m/s (stops)	-1.50 m/s ²	not given, not asked	? (wanted)

Step 3 Pick the equation.

AP sheet	equation	leaves out
#1	$v_x = v_{x0} + a_x t$	$x - x_0$ (Δx)
#2	$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	v_x (final velocity)
#3	$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	t (time) ← use this one

Why #3: time is not given and not asked for. #3, written $v_f^2 = v_i^2 + 2a\Delta x$, is the only one without t .

Step 4 Start from $v_f^2 = v_i^2 + 2a\Delta x$ and set $v_f = 0$. Solve for Δx before any numbers.

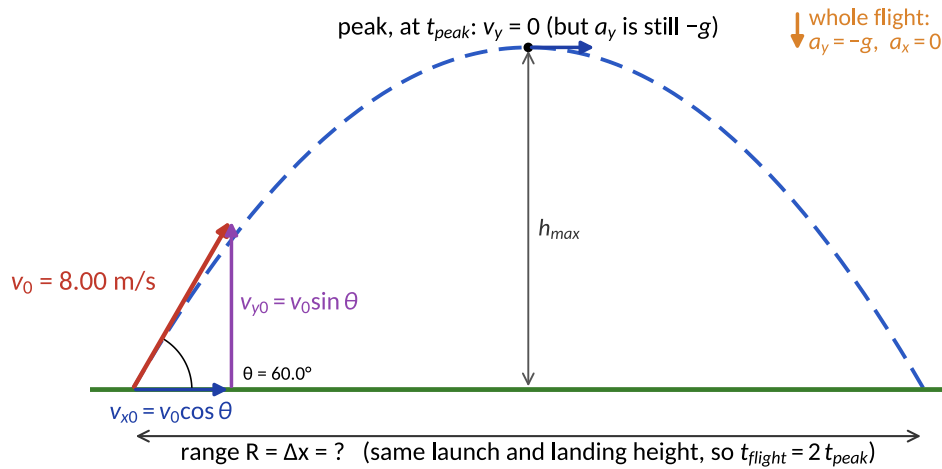
$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow 0 = v_i^2 + 2a\Delta x \rightarrow 2a\Delta x = -v_i^2 \rightarrow \Delta x = -\frac{v_i^2}{2a}$$

Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+27.0 \text{ m/s})^2}{2(-1.50 \text{ m/s}^2)} = -\frac{(729 \text{ m}^2/\text{s}^2)}{(-3.00 \text{ m/s}^2)} = \boxed{243 \text{ m}}$$

The minus sign in the formula and the negative a cancel, so Δx comes out positive: the train moves forward while it stops. If you get a negative distance, a had the wrong sign. Units: $(\text{m}^2/\text{s}^2) \div (\text{m}/\text{s}^2) = \text{m}$.

3. Sprinkler: range (25 points)



Split the motion into two motions. Sideways moves at a constant speed ($a_x = 0$). Up and down is free fall ($a_y = -g$). **Time** is the only thing both share.

Step 1 Resolve the initial velocity into components (keep letters; numbers go in at the end).

$v_{x0} = v_0 \cos \theta$ (the side next to the angle: along the ground)

$v_{y0} = v_0 \sin \theta$ (the side across from the angle: straight up)

Step 2 Make the T-chart.

horizontal	vertical
$a_x = 0$	$a_y = -g = -9.80 \text{ m/s}^2$
$v_{x0} = v_0 \cos \theta$	$v_{y0} = v_0 \sin \theta$
$\Delta x = R = ?$	at the peak: $v_y = 0$
$t = t_{\text{flight}}$	$t_{\text{peak}} = ?$

Step 3 Find the time to the peak, where $v_y = 0$.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0 (\Delta y) \leftarrow \text{use this one}$
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2 a_y (y - y_0)$	t (time)

Why #1: at the peak you know $v_y = 0$, v_{y0} and a_y , and you want t . The height of the peak is not known, so use #1, the one that leaves out Δy .

$$v_y = v_{y0} + a_y t \rightarrow 0 = v_0 \sin \theta - g t_{\text{peak}} \rightarrow t_{\text{peak}} = \frac{v_0 \sin \theta}{g}$$

Step 4 Double it for the whole flight.

The water lands at the same height it left the sprinkler, so the trip down takes exactly as long as the trip up: $t_{\text{flight}} = 2t_{\text{peak}} = \frac{2v_0 \sin \theta}{g}$

Watch out: at the peak $v_y = 0$, but the acceleration is still $a_y = -g$, pointing down. Gravity never switches off. Doubling t_{peak} only works because the launch and landing heights match. And $\Delta y = 0$ over the whole flight means it ends at the height it started, not that the acceleration is zero.

Step 5 Use the flight time in the horizontal column.

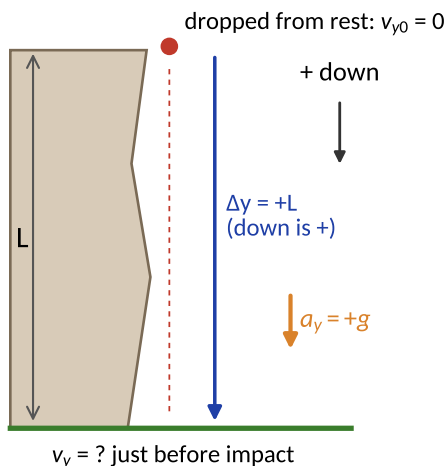
Why #2 (x-form): horizontally you know t and v_{x0} and want Δx ; with $a_x = 0$, #2 becomes $\Delta x = v_{x0}t$.

$$x - x_0 = v_{x0}t + \frac{1}{2}a_x t^2 \rightarrow R = v_{x0}t_{\text{flight}} = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} \rightarrow \boxed{R = \frac{2v_0^2 \sin \theta \cos \theta}{g}}$$

Step 6 Now plug in numbers with units.

$$R = \frac{2(8.00 \text{ m/s})^2(\sin 60.0^\circ)(\cos 60.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{5.66 \text{ m}}$$

Check: $t_{\text{peak}} = (8.00 \text{ m/s})(\sin 60.0^\circ)/(9.80 \text{ m/s}^2) = 0.7070 \text{ s}$, $t_{\text{flight}} = 1.414 \text{ s}$, $v_{x0} = 4.000 \text{ m/s}$, and $(4.000 \text{ m/s})(1.414 \text{ s}) = 5.656 \text{ m} \approx 5.66 \text{ m}$. ✓ (2 sin θ cos θ = sin 2θ , so this also equals $v_0^2 \sin 2\theta / g$.)

4. Carabiner dropped off a cliff: impact velocity (25 points), variables only

No numbers anywhere: the answer is in terms of L and g , both positive numbers.

This time **down is +** (the question says so), so the signs flip from the usual setup. The carabiner ends L below the start, which is the + direction: $\Delta y = +L$. Gravity also points down: $a_y = +g$.

Step 1 List the five variables (vertical, so y in place of x).

v_{y0} (initial)	v_y (final)	a_y	t	$y - y_0 = \Delta y$
0 (dropped from rest)	? (wanted)	$+g$	not given, not asked	$+L$

Step 2 Pick the equation.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time) ← use this one

Why #3: t is not given and not asked, and you want the final velocity. #3 is the only one without t.

Step 3 Solve with letters.

$$v_y^2 = v_{y0}^2 + 2a_y(y - y_0) = 0^2 + 2(+g)(+L) = 2gL \rightarrow v_y = \pm\sqrt{2gL}$$

Step 4 Pick the root.

It is moving **down** at the ground, and down is + here, so take the positive root: $v_y = +\sqrt{2gL}$

Speed = $\sqrt{2gL}$, same as with up as +; only the sign follows the convention.

Units check: $\sqrt{(m/s^2)(m)} = \sqrt{m^2/s^2} = m/s$.

Why not $-\sqrt{2gL}$? With down as +, that would mean moving up at the ground. Don't add a minus just because things fall; the sign comes from the convention.

Optional extra practice (not scored)**E1. Sprinkler: time in the air and maximum height**

$$(a) \text{ Time in the air } = t_{\text{flight}} = \frac{2v_0 \sin \theta}{g} = \frac{2(8.00 \text{ m/s})(\sin 60.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{1.41 \text{ s}}$$

Why #3 (y-form) for (b): at the peak $v_y = 0$, and you want the height without needing t.

$$(b) 0 = (v_0 \sin \theta)^2 - 2g h_{\text{max}} \rightarrow h_{\text{max}} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{(8.00 \text{ m/s})^2 (\sin 60.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{2.45 \text{ m}}$$

E2. Braking train: stopping time and the doubling question

Why #1: now t is wanted and Δx is not needed. $v_f = v_i + at \rightarrow 0 = v_i + at \rightarrow t = -\frac{v_i}{a} = -\frac{(+27.0 \text{ m/s})}{(-1.50 \text{ m/s}^2)} =$

$$\boxed{18.0 \text{ s}}$$

$$(b) \Delta x = -\frac{v_i^2}{2a} : v_i \text{ is squared, so doubling it multiplies } \Delta x \text{ by } 2^2 = \boxed{4} (4 \times 243 \text{ m} = 972 \text{ m}). \text{ Not 2.}$$

E3. Carabiner dropped off a cliff: velocity as a function of time

Why #1 (y-form): it links v_y and t directly. Down is +. $v_y = 0 + (+g)t \rightarrow \boxed{v_y(t) = +gt}$

Why #2 (y-form) for the impact time: you know $\Delta y = +L$ and want t. $+L = \frac{1}{2}g t_f^2 \rightarrow t_f = \sqrt{2L/g}$, so it is valid for $\boxed{0 \leq t \leq \sqrt{2L/g}}$. Check: $g\sqrt{2L/g} = \sqrt{2gL}$, matching Question 4.

Suggested partial credit (25 points per question, 100 total)

1. Acceleration-time graph

Suggested partial credit	pts
Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each)	8
Correct slope value with sign and units for each segment (1 each)	4
Four horizontal segments at the correct accelerations (2 each)	8
Changes in acceleration occur at the correct times; no slanted lines	2
Axes labeled with quantity and units; readable scale	3
Total	25

2. Braking train

Suggested partial credit	pts
Starts from $v_f^2 = v_i^2 + 2a\Delta x$ (sheet #3) because time is not involved	5
Sets $v_f = 0$	3
Acceleration negative, with a reason (opposite the motion / slowing down)	4
Solves $\Delta x = -v_i^2 / (2a)$ before substituting	6
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

3. Sprinkler range

Suggested partial credit	pts
Resolves v_0 into $v_0 \cos \theta$ and $v_0 \sin \theta$	4
Time to peak from $v_y = 0$ with sheet #1: $t_{\text{peak}} = v_0 \sin \theta / g$	5
Doubles t_{peak} for the flight time, with the equal-height reason	4
$R = v_{x0}t$ with the expression for t substituted (not a rounded number)	5
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

4. Dropped carabiner

Suggested partial credit	pts
States $v_{y0} = 0$, $a_y = +g$ and $\Delta y = +L$, consistent with down positive	6
Uses sheet #3 (y-form), since t is not involved	4
Algebra to $v_y^2 = 2gL$	5
Chooses the positive root with a reason (moving down, and down is +)	6
Final $v_y = +\sqrt{2gL}$ boxed, and speed $\sqrt{2gL}$ distinguished	4
Total	25